EgoLife: Towards Egocentric Life Assistant

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Abstract

*Building a persistent, personalized life assistant from egocentric video requires datasets and methods that span far beyond the short, single-activity clips studied today. We introduce **EgoLife**, a 266-hour, week-long, multimodal egocentric dataset recorded by six participants cohabiting in an instrumented environment over seven consecutive days. The dataset captures synchronized ego and exo video, audio, IMU, and gaze signals, accompanied by over 400K dense visual-audio captions and speaker-diarized transcripts. On top of EgoLife, we propose **EgoLifeQA**, a benchmark of 3,000 long-context multiple-choice questions spanning five task categories—EntityLog, EventRecall, HabitInsight, RelationMap, and TaskMaster—of which 2,003 require more than two hours of contextual evidence. To tackle this challenge, we present **EgoButler**, a two-stage system comprising EgoGPT, an omni-modal video-language model instruction-tuned on 99K egocentric QA pairs, and EgoRAG, a hierarchical retrieval module that constructs clip-, hour-, and day-level memory summaries. EgoGPT achieves state-of-the-art results on EgoSchema (75.4), EgoThink (61.4), and EgoPlan (33.4). On EgoLifeQA, personalization improves the average accuracy from 33.1 to 36.0, and EgoRAG boosts performance on questions requiring over 24 hours of context from 25.0 to 35.4. Dataset, benchmark, and models are publicly released to advance research toward practical egocentric life assistants.*

1. Introduction

Imagine an AI assistant that remembers your entire week—the meals you prepared, the conversations you had, and the habits you formed—all from your own first-person perspective. Such an egocentric life assistant could answer questions like "How many times did I exercise this week?" or "Who did I talk to about the project on Tuesday?", offering genuinely personalized support grounded in continuous visual-audio experience. Achieving this vision demands both a dataset that captures the richness of daily life over extended periods and a method that can reason across hundreds of hours of multimodal data.

Existing egocentric datasets have driven substantial progress in first-person video understanding, yet they remain limited in temporal scope and social complexity. EPIC-KITCHENS [8] provides fine-grained kitchen activity annotations but is restricted to a single domain. Ego4D [10] offers 3,670 hours of diverse daily-life footage, though its clips average only 23 minutes and lack interpersonal dynamics. EgoExo4D [11] introduces multiview ego-exo pairs for skilled activities but focuses on short procedural tasks. None of these datasets support the kind of ultra-long, multi-person reasoning that a practical life assistant requires—questions that demand integrating evidence from days of continuous recording across multiple individuals sharing the same environment.

To bridge this gap, we present **EgoLife**, a week-long, multimodal, ego-exo egocentric dataset in which six participants cohabit an instrumented house ("EgoHouse") for seven consecutive days. Participants wear Meta Aria glasses capturing video, audio, IMU, and gaze signals, while 15 exocentric GoPro cameras and 2 mmWave devices provide complementary viewpoints and spatial sensing. The resulting 266 hours of synchronized footage are accompanied by over 400K dense visual-audio captions and speaker-diarized transcripts, enabling research on long-horizon, socially grounded egocentric understanding (Figure 1).

Building on this dataset, we introduce **EgoLifeQA**, a benchmark of 3,000 multiple-choice questions across five categories that test progressively deeper temporal reasoning: EntityLog (object tracking), EventRecall (episode retrieval), HabitInsight (pattern discovery), RelationMap (social dynamics), and TaskMaster (procedural planning). Crucially, 2,003 of 3,000 questions have a certificate length exceeding two hours, requiring models to integrate evidence scattered across days of recording.

To address this challenge, we propose **EgoButler**, a two-stage egocentric life assistant comprising EgoGPT and EgoRAG. EgoGPT extends LLaVA-OneVision [14] with an audio branch and is instruction-tuned on EgoIT-99K, a new collection of 99K egocentric QA pairs, with Day-1 personalization to learn participant identities. EgoRAG complements EgoGPT with hierarchical retrieval over clip-, hour-, and day-level summaries, enabling efficient evidence gathering across the full week of recordings.

Our main contributions are:

- **EgoLife**: a 266-hour, week-long, multimodal, ego-exo

egocentric dataset featuring six participants, dense visual-audio captions, and rich interpersonal interactions.

- **EgoLifeQA**: a benchmark of 3,000 long-context multiple-choice questions across five task categories, with certificate-length metadata quantifying the temporal scope required for each question.
- **EgoButler**: a two-stage system combining EgoGPT (omni-modal clip understanding with personalization) and EgoRAG (hierarchical retrieval), together with the EgoIT-99K instruction-tuning dataset.
- Comprehensive analysis demonstrating that personalization, caption quality, and hierarchical retrieval are each key drivers of long-horizon egocentric QA performance.

2. Related Work

2.1. Egocentric Video Datasets

Egocentric video datasets have progressed from domain-specific collections to large-scale, multimodal benchmarks. EPIC-KITCHENS [8] pioneered fine-grained action annotation in kitchen environments but is limited to a single activity domain with clips averaging 8.5 minutes. Ego4D [10] dramatically expanded the scale to 3,670 hours of daily-life footage captured by 931 participants worldwide, introducing benchmarks for episodic memory, hand-object interaction, and social understanding. EgoExo4D [11] paired egocentric and exocentric viewpoints for 1,286 hours of skilled activities, enabling cross-view learning but focusing on short procedural tasks. EgoExoLearn [12] further explored ego-exo alignment for task execution across 120 hours. EgoPet [3] studied animal-perspective activities from egocentric footage.

Despite this progress, existing datasets share two key limitations: individual recording sessions are short (minutes to at most a few hours), and they capture isolated activities rather than continuous, multi-person daily life. As shown in Table 1, EgoLife is the first egocentric dataset to provide ultra-long per-participant recordings (44.3 hours each) with rich interpersonal dynamics, multimodal sensing, and over 400K dense captions—enabling research on life-scale temporal reasoning that prior datasets cannot support.

2.2. Egocentric Video Benchmarks

Benchmarks for egocentric video understanding have grown in sophistication alongside dataset advances. EgoSchema [17] introduced 5,063 three-minute temporal reasoning questions derived from Ego4D, establishing the certificate-length framework for measuring temporal grounding difficulty. EgoPlan-Bench [5] evaluates action planning from egocentric observations, while EgoThink [6] probes higher-order cognitive capabilities such as causal reasoning and spatial understanding. HourVideo [4] pushed toward longer contexts with questions over 20-minute to 2-hour Ego4D segments. EgoMemoria [28] further explored episodic memory retrieval from varied-length clips.

However, as summarized in Table 2, all existing benchmarks have certificate lengths under two hours, with no questions requiring cross-day reasoning. EgoLifeQA fills this gap: 2,003 of its 3,000 questions require more than two hours of evidence, and its five task categories—including habit discovery and social relationship analysis—test capabilities that emerge only at life-scale temporal horizons.

2.3. Video-Language Models

Recent video-language models have made rapid progress on egocentric understanding. LLaVA-OneVision [14] and LLaVA-Video [32] demonstrated strong visual instruction-following for video, while Qwen2-VL [7] and InternVideo2 [27] advanced multimodal video representation learning. Oryx [16] extended long-context video reasoning. Proprietary models such as GPT-4o [19] and Gemini-1.5-Pro [25] support long-context multimodal inputs but remain constrained to limited video lengths and lack personalization. Retrieval-augmented generation has emerged as a complementary strategy for extending effective context in video understanding. Our EgoButler system builds on these advances by combining an instruction-tuned omni-modal model (EgoGPT) with hierarchical retrieval (EgoRAG), specifically designed for week-long egocentric reasoning with participant-aware personalization.

3. The EgoLife Dataset & Benchmark

3.1. Data Collection

The EgoLife dataset was collected in a purpose-built instrumented apartment called the EgoHouse, designed to capture realistic multi-person daily life over an extended period. Six participants (P1–P6) cohabited in the EgoHouse for seven consecutive days, each recording approximately eight hours of first-person footage per day using Meta Aria glasses [9, 21]. These smart glasses provide synchronized ego video (two SLAM cameras and one RGB camera), binaural audio, IMU data, and eye-tracking signals. To capture complementary perspectives, we deployed 15 exocentric GoPro cameras covering common areas and two mmWave radar devices on the second floor for spatial sensing (Figure 2).

All devices are temporally synchronized using EgoSync, a synchronization framework that aligns all egocentric and exocentric recording devices to a unified timeline. The 3D spatial structure of the EgoHouse is reconstructed using Aria Multi-MPS [9], providing spatial context for localization and activity grounding. The dataset spans 266 hours of retained footage across six participants, with each participant contributing approximately 44.3 hours. Figure 3 shows the diverse range of daily activities captured throughout the week.

Figure 1. **The Overview of EgoLife Project.** EgoLife comprises a 266-hour, week-long, multimodal, multi-person egocentric dataset, a long-context QA benchmark (EgoLifeQA) covering five task categories, and EgoButler—an agentic life assistant built from EgoGPT and EgoRAG.

Table 1. **Related Work for EgoLife Dataset — Overview of Egocentric Datasets.** For Modality, **V** denotes video, **G** denotes gaze, **I** denotes IMU, **3D** denotes 3D scans. The EgoLife dataset stands out for its ultra-long egocentric footage and rich interpersonal interactions.

Benchmark	Domain	Modality	#Captions	Size (hrs)	#Clips	Dur./Clip	Multiview	Interpersonal Dynamics
EPIC-KITCHENS [8]	Kitchen	**V**	20K+	100	700	8.5 min	✗	✗
Ego4D [10]	Daily Activities	**V G I 3D**	3.85M	3,670	9,645	22.8 min	✗	✗
EgoExo4D [11]	Skilled Activities	**V G I 3D**	500K+	1,286	5,035	1 to 42 min	✓	✗
EgoExoLearn [12]	Task Execution	**V G I**	-	120	432	13.4 min	✓	✗
EgoPet [3]	Animal Actions	**V G I**	-	84	6,646	45.5 sec	✗	✗
EgoLife	Daily Life	**V G I 3D**	400K+	266	6	44.3 h	✓	✓

Table 2. **Related Work for EgoLifeQA Benchmark.** The EgoLifeQA dataset is distinguished by its ultra-long video footage and certificate length, facilitating novel tasks such as habit discovery and relational interaction pattern analysis (see Figure 5 for details). **Note on Dur./Clip:** A clip is defined as a session with narrative continuity. For the EgoLife dataset, this value is derived from 266 hours of retained footage distributed across six participants.

Dataset	Source	#QAs	Size (hrs)	#Clips	Dur./Clip	Certificate Length [17]	
						Below 2h	Over 2h
EgoSchema [17]	Ego4D	5,063	250	5,063	3 min	5,063	0
EgoPlan-Bench [5]	Ego4D & EpicKitchen	4,939	-	4,939	-	4,939	0
EgoThink [6]	Ego4D	700	-	595	-	700	0
EgoMemoria [28]	Ego4D	7,026	-	629	30 s to 1 h	7,026	0
HourVideo [4]	Ego4D	12,976	381	500	20 min to 2 h	12,976	0
EgoLifeQA	**EgoLife**	3,000	266	6	44.3 h	997	2,003

Participants conducted their daily lives naturally, engaging in cooking, dining, socializing, working, exercising, and relaxing. Recordings are primarily in Chinese, with all transcripts provided with English translations. To protect participant privacy, all faces and identifying information in released materials are anonymized using EgoBlur.

3.2. Data Processing Pipeline

The data processing pipeline transforms raw multimodal recordings into structured, annotated data suitable for training and evaluation (Figure 4).

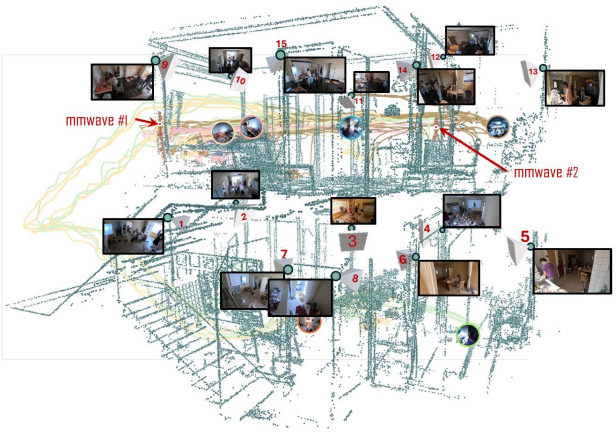


Figure 2. 3D reconstruction of the shared house using Aria Multi-MPS [9], showcasing the locations of 15 Exo cameras in the common area and 2 mmWave devices (highlighted in red) on the second floor.

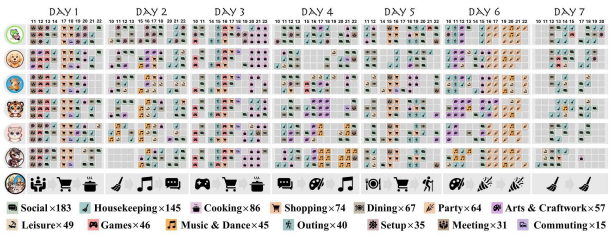


Figure 3. **The Activity Timeline of the EgoLife Dataset.** Daily activities of all six participants are illustrated over seven days of cohabitation, covering a broad spectrum of everyday life events.

Speech Transcription. Audio streams are transcribed using Whisper ASR [22], followed by speaker diarization [1] to assign utterances to individual participants. Human annotators review and correct all transcripts, producing time-aligned, speaker-attributed SRT files covering all recorded conversations.

Visual-Audio Caption Generation. Each recording session is segmented into five-minute clips. Human narrators watch each clip at 0.8× speed and record detailed audio descriptions of visual events alongside concurrent audio context from conversations. These per-clip narrations are then merged by GPT-4o [19] into dense, coherent visual-audio captions. This process yields over 400K captions providing rich textual grounding across the full dataset.

QA Construction and Quality Control. For each participant, approximately 100K candidate QA pairs are automatically generated from captions and transcripts using large language models. These are manually filtered and refined to 500 high-quality questions per participant (3,000 total). Each question includes carefully crafted distractors, audio-relevance flags indicating whether audio context is necessary for correct answering, and SRT-aligned temporal anchors. Certificate lengths are computed following [17] to quantify the minimum look-back duration required for correct answering.

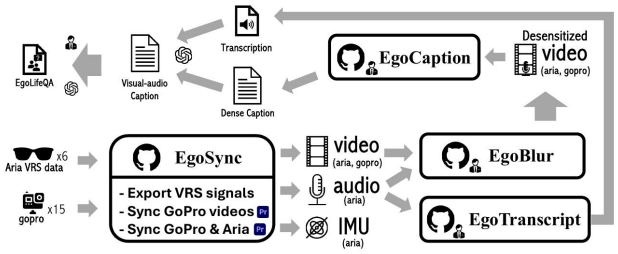


Figure 4. **The Overview of Data Process Pipeline.** Raw Aria and GoPro recordings are processed through transcription, privacy protection, caption generation, and QA construction to produce the EgoLife dataset and EgoLifeQA benchmark.

3.3. EgoLifeQA Benchmark

EgoLifeQA comprises 3,000 multiple-choice questions organized into five task categories that probe progressively deeper temporal reasoning capabilities (Figure 5):

- **EntityLog**: Track objects, items, or resources across time (e.g., "Where did I leave my keys on Tuesday?").
- **EventRecall**: Retrieve specific events from memory (e.g., "What did P3 cook for dinner on day 4?").
- **HabitInsight**: Discover recurring patterns and routines (e.g., "How often did I exercise in the morning this week?").
- **RelationMap**: Analyze interpersonal relationships and social dynamics (e.g., "Who most frequently joined P2 for breakfast?").
- **TaskMaster**: Track multi-step task progress and completion (e.g., "Did I complete all the tasks I planned on day 3?").

As shown in Figure 6 and Table 2, EgoLifeQA is uniquely characterized by its ultra-long certificate lengths: 2,003 of 3,000 questions (66.8%) require more than two hours of contextual evidence, compared to zero such questions in all existing egocentric benchmarks. This reflects the genuine demands of life-assistant tasks requiring reasoning across days of experience rather than isolated clips.

4. EgoButler: Agentic Egocentric Life Assistant

4.1. EgoGPT: Omni-Modal Clip Understanding

EgoGPT is a 7-billion-parameter omni-modal video-language model designed for first-person video understanding with synchronized audio. We extend LLaVA-OneVision [14] by adding a dedicated audio branch: speech features are extracted using a Whisper-large-v3 encoder and

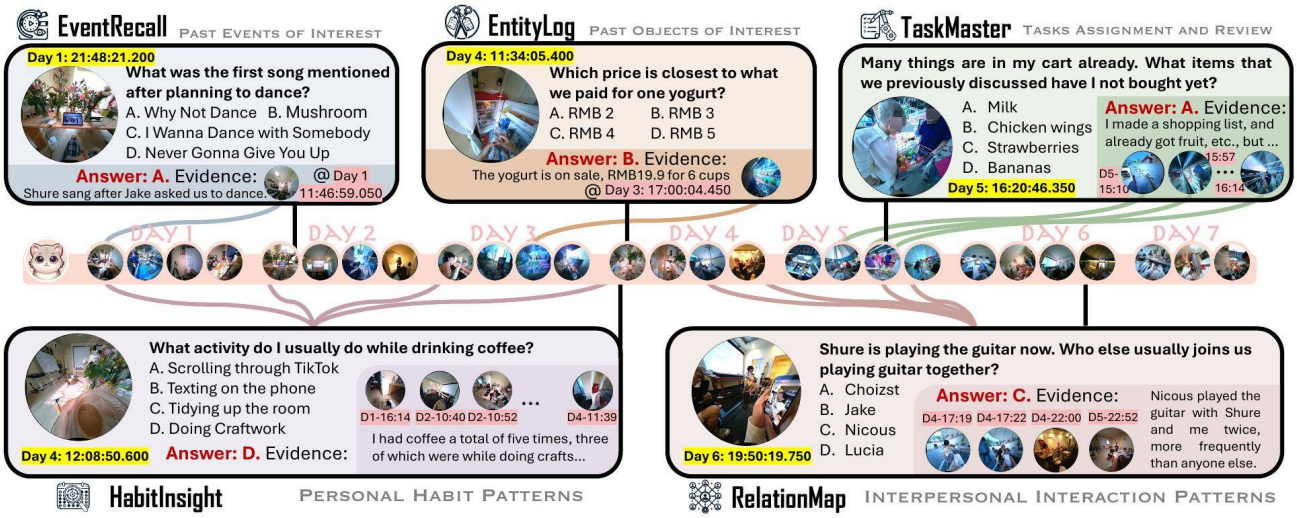


Figure 5. **Question Types and Examples in the EgoLifeQA Benchmark.** Five task categories are illustrated with representative questions and answer options, ranging from short-context entity tracking to long-horizon habit and relationship analysis requiring integration of days of evidence.

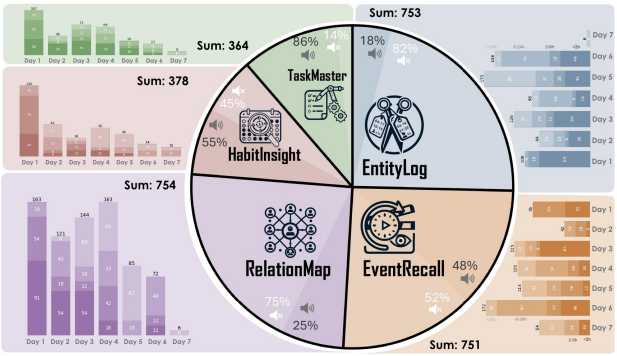


Figure 6. **Statistics of EgoLifeQA.** Distribution of certificate lengths and task categories across 3,000 QA pairs. The majority of questions require more than two hours of context, highlighting the unique long-horizon challenge of the benchmark.

projected into the language model embedding space via a lightweight MLP speech projector. Visual features follow the standard LLaVA-OV pathway with SigLIP visual encoding and an MLP projector. The combined visual and audio token sequences are provided to the Qwen2 language backbone alongside text tokens, enabling simultaneous multimodal reasoning.

EgoIT-99K Instruction Tuning. We construct EgoIT-99K, a collection of 99,480 egocentric instruction-tuning samples drawn from nine established egocentric datasets: Ego4D [10], Charades-Ego [24], HoloAssist [26], EGTEA Gaze+ [15], IndustReal [23], EgoTaskQA [13], EgoProceL [2], Epic-Kitchens [8], and ADL [20]. The data spans multiple QA types including video captioning (VC), audio-video captioning (AVC), multiple-choice questions (MCQ), multi-round conversations (MRC), and image QA (IQA), as detailed in Table 3. EgoGPT is instruction-tuned using 32 frames per video clip.

Day-1 Personalization. To enable identity-aware reasoning, we further fine-tune EgoGPT on each participant's first-day recordings (EgoLifeD1). This personalization step teaches the model to recognize participant identities, associate them with names, and adapt to individual speaking styles and activity patterns—capabilities essential for RelationMap and EntityLog tasks that require tracking specific individuals across days of footage.

4.2. EgoRAG: Hierarchical Retrieval for Long-Context QA

EgoGPT operates at the clip level (5-minute windows) and cannot directly process a full week of footage within its context window. EgoRAG addresses this fundamental limitation by building a multi-level memory bank and retrieving relevant evidence for each query (Figure 7).

Hierarchical Memory Construction. EgoGPT generates dense captions for all five-minute clips in the dataset. These clip-level captions are aggregated into hour-level and day-level summaries using GPT-4o, creating a three-level memory hierarchy that spans the full recording period. This hierarchical structure supports both fine-grained event retrieval and coarse-grained temporal pattern discovery.

Retrieval and Answer Generation. Given a test question, EgoRAG encodes the query using dense text embeddings and retrieves the top-k most relevant clip captions from the

Table 3. **Dataset Composition of EgoIT-99K.** We curated 9 classic egocentric video datasets and leveraged their annotations to generate captioning and QA instruction-tuning data for fine-tuning EgoGPT, building on the LLaVA-OneVision base model [14]. **#AV** means the number of videos with audio used for training. QAs include multiple types—**VC**: Video Captioning, **AVC**: Audio-Video Captioning, **MCQ**: Multiple Choice Questions, **MRC**: Multi-Round Questions, **IQA**: Image Question-Answering.

Dataset	Duration	#Videos (#AV)	#QA	QA Type
Ego4D [10]	3.34h	523 (458)	1.41K	VC, AVC, MCQ, MRC
Charades-Ego [24]	5.04h	591 (228)	18.46K	VC, AVC, MRC
HoloAssist [26]	9.17h	121	33.96K	VC, MCQ, MRC, IQA
EGTEA Gaze+ [15]	3.01h	16	11.20K	VC, MCQ, MRC, IQA
IndustReal [23]	2.96h	44	11.58K	VC, MCQ, MRC, IQA
EgoTaskQA [13]	8.72h	172	3.59K	VC, MCQ, MRC
EgoProceL [2]	3.11h	18	5.90K	VC, MCQ, MRC, IQA
Epic-Kitchens [8]	4.15h	36	10.15K	VC, MCQ, MRC, IQA
ADL [20]	3.66h	8	3.23K	VC, MCQ, MRC, IQA
Total	**43.16h**	**1529 (686)**	**99.48K**	

memory bank. The retrieved captions, together with the original question, are provided to EgoGPT as context for final answer generation. This design enables EgoButler to leverage evidence from across the full week of recordings while remaining computationally tractable at inference time.

5. Experiments

5.1. Implementation Details

EgoGPT is built on LLaVA-OneVision-7B [14] with a Whisper-large-v3 audio encoder and MLP speech projector. Fine-tuning uses the AdamW optimizer with a cosine learning rate schedule: first on EgoIT-99K for general egocentric instruction tuning, then on EgoLifeD1 for participant-specific personalization. All models use 32 frames per clip. For EgoLifeQA evaluation, video-based models sample at 1 FPS. EgoRAG constructs clip-level captions at 1 FPS and indexes them with dense text embeddings, retrieving the top-10 most relevant clips per question.

5.2. EgoGPT on Existing Benchmarks

Table 4 compares EgoGPT against state-of-the-art commercial and open-source video-language models on three established egocentric benchmarks: EgoSchema [17], EgoPlan [5], and EgoThink [6].

EgoGPT (EgoIT-99K) achieves 73.2 on EgoSchema, already surpassing all 7B open-source baselines including Qwen2-VL [7] (66.7), LLaVA-OV [14] (60.1), and InternVideo2 [27] (55.2). With Day-1 personalization, EgoGPT (EgoIT-99K+D1) further improves to 75.4 on EgoSchema—surpassing Gemini-1.5-Pro [25] (72.2) and GPT-4o [19] (72.2)—while achieving 33.4 on EgoPlan and 61.4 on EgoThink. These results demonstrate that egocentric-specific instruction tuning and personalization yield consistent gains across multiple benchmarks.

Table 4. **Performance of EgoGPT.** The table compares EgoGPT with state-of-the-art commercial and open-source models on existing egocentric benchmarks.

Model	#Param	#Frames	EgoSchema	EgoPlan	EgoThink
GPT-4v [18]	-	32	56.6	38.0	65.5
Gemini-1.5-Pro [25]	-	32	72.2	31.3	62.4
GPT-4o [19]	-	32	72.2	32.8	65.5
LLaVA-Next-Video [31]	7B	32	49.7	29.0	40.6
LongVA [30]	7B	32	44.1	29.9	48.3
IXC-2.5 [29]	7B	32	54.6	29.4	56.0
InternVideo2 [27]	8B	32	55.2	27.5	43.9
Qwen2-VL [7]	7B	32	66.7	34.3	59.3
Oryx [16]	7B	32	56.0	33.2	53.1
LLaVA-OV [14]	7B	32	60.1	30.7	54.2
LLaVA-Videos [32]	7B	32	57.3	33.6	56.4
EgoGPT (EgoIT)	7B	32	73.2	32.4	61.7
EgoGPT (EgoIT+EgoLifeD1)	7B	32	75.4	33.4	61.4

5.3. EgoButler on EgoLifeQA

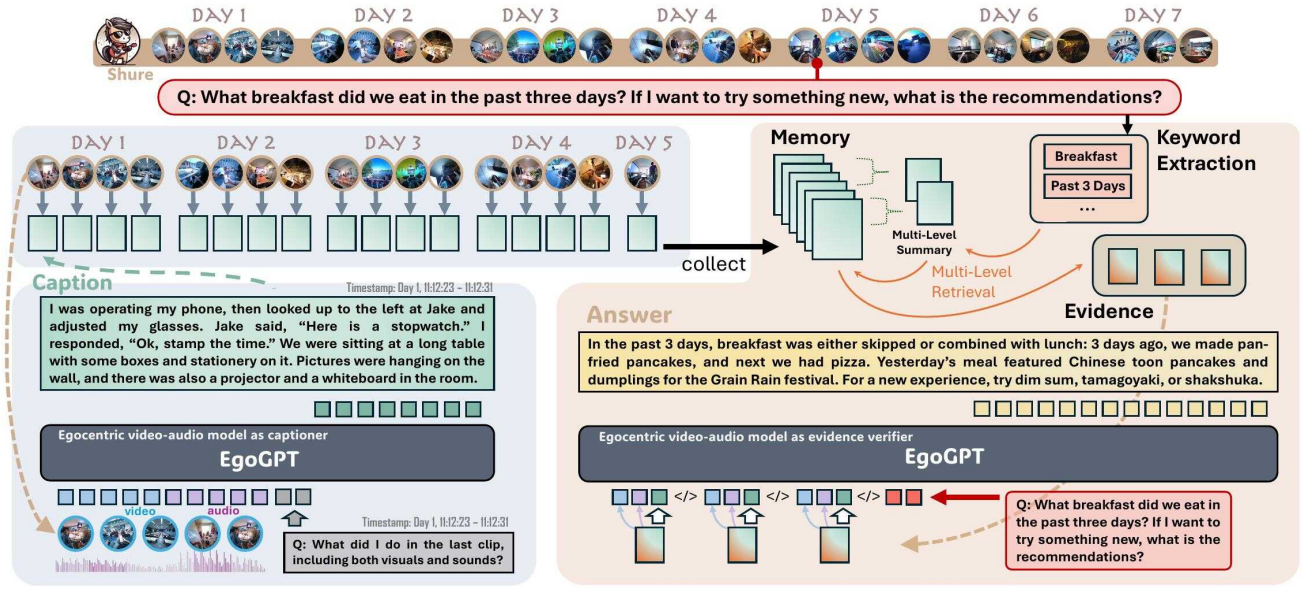
Table 5 compares EgoButler variants with commercial and open-source baselines on EgoLifeQA. For a fair comparison, each baseline model replaces EgoGPT within the EgoButler pipeline, and video-based models use 1 FPS caption sampling.

EgoGPT (EgoIT-99K) achieves 33.1 average accuracy, surpassing LLaVA-OV (30.8) while remaining below Gemini-1.5-Pro (36.9) and GPT-4o (36.2), both of which benefit from larger model capacity. Adding Day-1 personalization (EgoIT-99K+D1) improves the average to 36.0, with particularly large gains on EntityLog (+4.0, from 35.2 to 39.2) and RelationMap (+4.0, from 29.6 to 33.6)—tasks that explicitly require recognizing participant identities. This confirms that personalization is a critical component for socially grounded egocentric reasoning.

5.4. Ablation Studies

EgoRAG Effectiveness. Table 6 analyzes EgoRAG performance decomposed by certificate length. Without retrieval, EgoGPT achieves only 26.8 accuracy on 6–24h questions and 25.0 on >24h questions—barely above random chance for a four-choice task. Adding EgoRAG dramatically improves these to 38.9 (+12.1) and 35.4 (+10.4) respectively, while maintaining comparable performance on short-context questions (<2h: 27.2 vs. 28.2). Notably, Gemini-1.5-Pro without retrieval scores even lower at 25.0 for 6–24h and 18.4 for >24h, underscoring that raw model capacity alone cannot substitute for explicit temporal evidence retrieval in ultra-long-context settings.

Caption Quality Ablation. Table 7 examines how different caption sources affect EgoButler performance. Human visual-audio captions serve as an upper bound at 45.5 average accuracy, significantly above all automated variants. Among automatic caption sources, audio-only EgoGPT yields 27.2, visual-only reaches 31.2, and combined visual-audio achieves 33.1—demonstrating that the audio modality contributes meaningful and complementary information beyond visual content alone. Day-1 personalization consistently improves each configuration, with visual+audio+D1



(a) Captioning Stage with EgoGPT **(b) Question Answering Stage with EgoRAG**

Figure 7. **The EgoButler Architecture.** EgoButler integrates EgoGPT (an omni-modal video-language model with audio branch and Day-1 personalization) and EgoRAG (a hierarchical retrieval system over clip-, hour-, and day-level memory summaries) to answer week-long egocentric life queries.

Table 5. **Performance comparison of EgoGPT with state-of-the-art models on EgoLifeQA benchmarks.** For a fair comparison on EgoLifeQA, EgoGPT was replaced with the corresponding models in the EgoButler pipeline to evaluate their performance under the same conditions. Models that provide captions for EgoLifeQA use 1 FPS for video sampling.

Model	#Frames	Audio	Identity	EgoLifeQA					
				EntityLog	EventRecall	HabitInsight	RelationMap	TaskMaster	Average
Gemini-1.5-Pro [25]	-	✓	✗	36.0	37.3	45.9	30.4	34.9	36.9
GPT-4o [19]	1 FPS	✗	✗	34.4	42.1	29.5	30.4	44.4	36.2
LLaVA-OV [14]	1 FPS	✗	✗	36.8	34.9	31.1	22.4	28.6	30.8
EgoGPT (EgoIT-99K)	1 FPS	✓	✗	35.2	36.5	27.9	29.6	36.5	33.1
EgoGPT (EgoIT-99K+D1)	1 FPS	✓	✓	39.2	36.5	31.1	33.6	39.7	36.0

Table 6. **Effectiveness of EgoRAG.** Integrating EgoRAG significantly enhances video-language models' performance in long-context question answering, especially for questions requiring longer certification lengths. For comparison, we evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment, limiting their answers to this timeframe.

Model	Certificate Length			
	$< 2h$	$2h - 6h$	$6h - 24h$	$> 24h$
Gemini-1.5-Pro	27.9	14.8	25.0	18.4
EgoGPT	28.2	29.1	26.8	25.0
EgoGPT+EgoRAG	27.2	35.7	38.9	35.4

reaching 36.0. The substantial gap between the automated best (36.0) and the human ceiling (45.5) highlights the remaining challenge in automated egocentric caption generation and motivates future research in this direction.

Qualitative Analysis. Figure 8 presents representative qualitative examples comparing EgoGPT and Gemini-1.5-Pro under the EgoButler framework. EgoGPT with Day-1 personalization correctly identifies participants by name and

Table 7. **Ablation Study on EgoGPT.** We construct different EgoRAG memory banks using generated captions from EgoGPT variants. The first three rows use captions from human annotations as a reference. All response generation models utilize EgoGPT (EgoIT-99K+D1) to ensure fair comparison. The result indicates how caption quality affects of EgoBulter performance.

Caption Source	Visual	Audio	Dataset	Avg.
Narration	✓	✓	-	31.5
Transcript	✗	✓	-	29.6
Visual-Audio Caption	✓	✓	-	45.5
EgoGPT (Audio Only)	✗	✓	EgoIT-99K	27.2
EgoGPT (Audio Only)	✗	✓	EgoIT-99K+D1	28.1
EgoGPT (Visual Only)	✓	✗	EgoIT-99K	31.2
EgoGPT (Visual Only)	✓	✗	EgoIT-99K+D1	33.6
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K	33.1
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K+D1	36.0

retrieves temporally relevant evidence, correctly answering questions about social interactions and personal habits. In contrast, Gemini-1.5-Pro without identity awareness fails on RelationMap and EntityLog questions that require associating actions with specific individuals.

6. Conclusion and Outlook

We have presented EgoLife, EgoLifeQA, and EgoButler—a comprehensive framework for advancing egocentric life assistance. EgoLife provides 266 hours of synchronized multimodal data from six cohabiting participants over seven days, with over 400K dense visual-audio captions and speaker-attributed transcripts, establishing a new scale for interpersonal, long-horizon egocentric research. EgoLifeQA introduces 3,000 long-context multiple-choice questions across five task categories, of which 2,003 require more than two hours of contextual evidence—setting new challenges in ultra-long temporal reasoning that prior benchmarks do not address. EgoButler combines EgoGPT, an omni-modal model instruction-tuned on EgoIT-99K with participant-specific personalization, and EgoRAG for hierarchical evidence retrieval across week-long recordings. Our experiments demonstrate that personalization, caption quality, and hierarchical retrieval are each independently critical for performance, and together enable state-of-the-art results on both existing egocentric benchmarks and EgoLifeQA. We release all dataset, benchmark, and model components to support the community in building practical egocentric life assistants.

References

[1] Mahmoud Ashraf. Whisper diarization: Speaker diarization using openai whisper. 2024. 4

[2] Siddhant Bansal, Chetan Arora, and C.V. Jawahar. My view is the best view: Procedure learning from egocentric videos. In European Conference on Computer Vision (ECCV), 2022. 5, 6

[3] Amir Bar, Arya Bakhtiar, Danny Tran, Antonio Loquercio, Jathushan Rajasegaran, Yann LeCun, Amir Globerson, and Trevor Darrell. Egopet: Egomotion and interaction data from an animal's perspective. arXiv preprint arXiv:2404.09991, 2024. 2, 3

[4] Keshigeyan Chandrasegaran, Agrim Gupta, Lea M. Hadzic, Taran Kota, Jimming He, Cristóbal Eyzaguirre, Zane Durante, Manling Li, Jiajun Wu, and Li Fei-Fei. Hourvideo: 1-hour video-language understanding, 2024. 2, 3

[5] Yi Chen, Yuying Ge, Yixiao Ge, Mingyu Ding, Bohao Li, Rui Wang, Ruifeng Xu, Ying Shan, and Xihui Liu. Egoplan-bench: Benchmarking egocentric embodied planning with multimodal large language models. arXiv preprint arXiv:2312.06722, 2023. 2, 3, 6

[6] Sijie Cheng, Zhicheng Guo, Jingwen Wu, Kechen Fang, Peng Li, Huaping Liu, and Yang Liu. Egothink: Evaluating first-person perspective thinking capability of vision-language models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 14291–14302, 2024. 2, 3, 6

[7] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, et al. Qwen2-audio technical report. arXiv preprint arXiv:2407.10759, 2024. 2, 6

[8] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, et al. The epic-kitchens dataset: Collection, challenges and baselines. IEEE Transactions on Pattern Analysis and Machine Intelligence, 43(11): 4125–4141, 2021. 1, 2, 3, 5, 6

[9] Jakob Engel, Kiran Somasundaram, Michael Goesele, Albert Sun, Alexander Gamino, Andrew Turner, Arjang Talattof, Arnie Yuan, Bilal Souti, Brighid Meredith, et al. Project aria: A new tool for egocentric multimodal ai research. arXiv preprint arXiv:2308.13561, 2023. 2, 4

[10] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, et al. Ego4d: Around the world in 3,000 hours of egocentric video. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 18995–19012, 2022. 1, 2, 3, 5, 6

[11] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, et al. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 19383–19400, 2024. 1, 2, 3

[12] Yifei Huang, Guo Chen, Jilan Xu, Mingfang Zhang, Lijin Yang, Baoqi Pei, Hongjie Zhang, Dong Lu, Yali Wang, et al. Egoexolearn: A dataset for bridging asynchronous ego- and exo-centric view of procedural activities in real world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. 2, 3

[13] Baoxiong Jia, Ting Lei, Song-Chun Zhu, and Siyuan Huang. Egotaskqa: Understanding human tasks in egocentric videos. Advances in Neural Information Processing Systems, 35:3343–3360, 2022. 5, 6

[14] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task

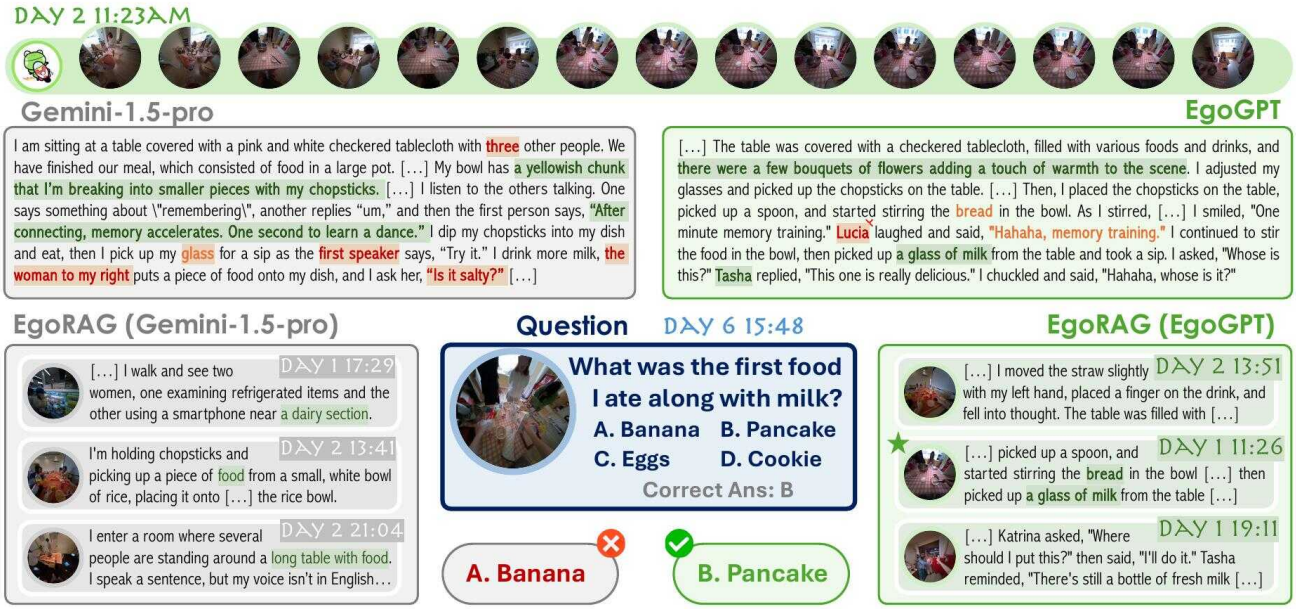


Figure 8. **Qualitative Comparison of EgoGPT and Gemini-1.5-Pro under the EgoButler Framework.** EgoGPT with Day-1 personalization correctly identifies participants and retrieves relevant temporal evidence for long-horizon questions. Gemini-1.5-Pro without identity awareness struggles with socially grounded queries.

transfer. arXiv preprint arXiv:2408.03326, 2024. 1, 2, 4, 6, 7

[15] Yin Li, Miao Liu, and James M. Rehg. In the eye of beholder: Joint learning of gaze and actions in first person video. In European Conference on Computer Vision (ECCV), 2018. 5, 6

[16] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx mllm: On-demand spatial-temporal understanding at arbitrary resolution. arXiv preprint arXiv:2409.12961, 2024. 2, 6

[17] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. In Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2023. 2, 3, 4, 6

[18] OpenAI. Gpt-4v(ision) system card, 2023. 6

[19] OpenAI. Gpt-4o system card, 2024. 2, 4, 6, 7

[20] Junhao Pan, Zehua Yuan, Xiaofan Zhang, and Deming Chen. Youhome system and dataset: Making your home know you better. IEEE International Symposium on Smart Electronic Systems (IEEE - iSES), 2022. 5, 6

[21] Xiaqing Pan, Nicholas Charron, Yongqian Yang, Scott Peters, Thomas Whelan, Chen Kong, Omkar Parkhi, Richard Newcombe, and Yuheng Carl Ren. Aria digital twin: A new benchmark dataset for egocentric 3d machine perception. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 20133–20143, 2023. 2

[22] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In International conference on machine learning, pages 28492–28518. PMLR, 2023. 4

[23] Tim J Schoonbeek, Tim Houben, Hans Onvlee, Fons van der Sommen, et al. Industreal: A dataset for procedure step recognition handling execution errors in egocentric videos in an industrial-like setting. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pages 4365–4374, 2024. 5, 6

[24] Gunnar A. Sigurdsson, Abhinav Gupta, Cordelia Schmid, Ali Farhadi, and Karteek Alahari. Charades-ego: A large-scale dataset of paired third and first person videos, 2018. 5, 6

[25] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. arXiv preprint arXiv:2312.11805, 2023. 2, 6, 7

[26] Xin Wang, Taein Kwon, Mahdi Rad, Bowen Pan, Ishani Chakraborty, Sean Andrist, Dan Bohus, Ashley Feniello, Bugra Tekin, Felipe Vieira Frujeri, et al. Holoassist: an egocentric human interaction dataset for inter-

active ai assistants in the real world. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 20270–20281, 2023. 5, 6

[27] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yinan He, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang, et al. Internvideo2: Scaling video foundation models for multimodal video understanding. arXiv preprint arXiv:2403.15377, 2024. 2, 6

[28] Hanrong Ye, Haotian Zhang, Erik Daxberger, Lin Chen, Zongyu Lin, Yanghao Li, Bowen Zhang, Haoxuan You, Dan Xu, Zhe Gan, et al. Mm-ego: Towards building egocentric multimodal llms. arXiv preprint arXiv:2410.07177, 2024. 2, 3

[29] Pan Zhang, Xiaoyi Dong, Bin Wang, Yuhang Cao, Chao Xu, Linke Ouyang, Zhiyuan Zhao, Haodong Duan, Songyang Zhang, Shuangrui Ding, et al. Internlm-xcomposer: A vision-language large model for advanced text-image comprehension and composition. arXiv preprint arXiv:2309.15112, 2023. 6

[30] Peiyuan Zhang, Kaichen Zhang, Bo Li, Guangtao Zeng, Jingkang Yang, Yuanhan Zhang, Ziyue Wang, Haoran Tan, Chunyuan Li, and Ziwei Liu. Long context transfer from language to vision. arXiv preprint arXiv:2406.16852, 2024. 6

[31] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model, 2024. 6

[32] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data. arXiv preprint arXiv:2410.02713, 2024. 2, 6